

Comprehensive RAG Comparison Report

QuitTxt Protocol-Guided AI Smoking Cessation System

REVISED - Rigorous Methodology

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1 Executive Summary

This report presents **rigorously validated** results using fair methodology. All approaches were tested under **identical conditions**: same 15 test questions, same LLM (GPT-4o-mini), same retrieval method (ChromaDB with cosine similarity).

Key Results (5 runs, 95% CI):

- AI-Generated achieves **96.9%** of Human-Curated performance (ROUGE-L)
- Unexpectedly, **Raw Sources outperform processed datasets**
- Domain saturation effect: LLM already knows smoking cessation well

2 Methodology

2.1 Fair Comparison Protocol

- **Test Questions:** 15 held-out questions (same for all approaches)
- **Runs:** 5 independent trials for statistical significance
- **Retrieval:** ChromaDB with cosine similarity embeddings
- **LLM:** GPT-4o-mini (temperature=0.3)
- **Primary Metric:** ROUGE-L (semantic similarity)

2.2 Approaches Compared

1. **Baseline (No RAG)** – LLM answers from internal knowledge only
2. **Raw Sources RAG** – Using unprocessed web content snippets
3. **AI-Generated RAG** – Synthetic Q&A pairs from Gemini 2.0 Flash
4. **Human-Curated RAG** – Expert-created Q&A pairs (4,847 pairs)

Approach	Mean	95% CI	Rank
Raw Sources RAG	0.254	± 0.004	#1
Human-Curated RAG	0.244	± 0.004	#2
Baseline (No RAG)	0.244	± 0.005	#2 (tie)
AI-Generated RAG	0.237	± 0.007	#4

Table 1: ROUGE-L results showing Raw Sources performs best

3 Results

3.1 Primary Metric: ROUGE-L (Semantic Similarity)

3.2 Secondary Metric: BLEU (Lexical Similarity)

Approach	Mean	95% CI	Rank
Raw Sources RAG	0.100	± 0.003	#1
Human-Curated RAG	0.085	± 0.003	#2
AI-Generated RAG	0.081	± 0.005	#3
Baseline (No RAG)	0.071	± 0.004	#4

Table 2: BLEU results confirming Raw Sources performs best

3.3 Key Comparisons

Comparison	ROUGE-L	BLEU
AI % of Human	96.9%	95.4%
AI vs Raw	-6.8%	-19.2%
Human vs Raw	-3.8%	-15.3%

Table 3: AI achieves near-human performance, but both underperform Raw

4 Target Validation

5 Important Findings

5.1 Unexpected Result: Raw Outperforms Processed

The expected hierarchy was: Raw < AI < Human

The actual hierarchy is: **Raw** > **Human** $\approx$ **Baseline** > **AI**

Possible explanations:

- Q&A formatting may constrain retrieval
- Raw text provides more flexible context
- Domain is well-known to the LLM (saturation effect)

Target (from Meeting)	Expected	Actual	Status
AI achieves 84% of Human (ROUGE-L)	84%	96.9%	MET
AI 40% better than Raw	+40%	-6.8%	NOT MET

Table 4: Professor’s targets assessment

5.2 Vocabulary Alignment Hypothesis: NOT Supported

Dataset	Overlap with LLM
Human-curated	96.2%
AI-generated	92.2%

Table 5: Human content has MORE LLM vocabulary overlap

5.3 Methodological Note

Previous reports contained invalid comparisons. The earlier claimed improvements (149%, 254%) compared different experiments with different test questions. BLEU/ROUGE scores cannot be compared across different test sets.

6 Conclusions

1. **AI-Generated is viable:** Achieves **96.9%** of human-curated semantic performance
2. **Simpler may be better:** Raw sources unexpectedly outperform processed datasets
3. **Domain matters:** Smoking cessation is well-covered in LLM training data
4. **Methodology is critical:** Fair comparisons require identical test conditions
5. **Q&A formatting has costs:** Converting to Q&A format may lose information

7 What Can Be Claimed

Valid claims:

- AI-generated achieves near-human quality (97%)
- All approaches perform similarly for this domain
- Domain knowledge saturation reduces RAG benefit

Invalid claims (do not use):

- ~~Structured preprocessing improves performance~~
- ~~149% or 254% improvements~~
- ~~Human-curated is gold standard~~